

Advanced (Habanero)

- Q1.** What is the simplified ratio of $2 \cdot x$ to $6 \cdot x$?
(A) 1:2
(B) 1:3
(C) 2:3
(D) 3:6
(E) 1:6
- Q2.** If $\frac{a}{b} = \frac{2}{3}$, what is $\frac{3a}{2b}$ equal to?
(A) $\frac{1}{2}$
(B) $\frac{3}{2}$
(C) $\frac{9}{4}$
(D) 1
(E) 2
- Q3.** 5 apples to 3 oranges is the same ratio as x apples to 9 oranges. What is x ?
(A) 10
(B) 12
(C) 15
(D) 20
(E) 25
- Q4.** A recipe requires 2 cups of flour to 1 cup of sugar. How many cups of flour are required for 3 cups of sugar?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10
- Q5.** A car travels 250 miles in 5 hours. How many miles will it travel in 7 hours at the same rate?
(A) 300
(B) 350
(C) 400
(D) 450
(E) 500
- Q6.** A ratio of boys to girls in a school is 3 : 5. If there are 18 girls, how many boys are there?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 15
- Q7.** If $\frac{2}{x} = \frac{5}{12}$, what is x equal to?
(A) $\frac{5}{3}$
(B) $\frac{10}{3}$
(C) $\frac{15}{2}$
(D) $\frac{24}{5}$
(E) $\frac{36}{5}$
- Q8.** A map has a scale of 1 : 50,000. If two towns are 5 cm apart on the map, how far apart are they in reality?
(A) 200 m
(B) 250 m
(C) 250,000 m
(D) 50,000 cm
(E) 25 km

Open Ended Questions (FRQ)

- Q9.** A bakery sells 480 loaves of bread per day. They make a profit of \$1.20 per loaf. Express their daily profit as a ratio of their total daily revenue to their daily expenses, if their daily expenses are \$500.
- Q10.** In a contest, the ratio of winning to total tickets is 1 : 100. If 500 tickets are sold, what is the expected number of winning tickets? Show your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of simplifying ratios.
- Remind students that proportions can be used to model real-world situations.
- Encourage students to check their work by using inverse operations.